

Question Paper

Exam Date & Time: 21-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: ECE 3122/ECE_3122 MICROPROCESSORS

(SPL: ELECTRONICS & COMMUNICATION ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Using a neat timing diagram, show how many cycles it takes for a 3-stage pipelined ARM7 CPU to execute the following code. (4)

```
MOV R1, #0
MOV R4, #3
L1:  ADD R2, R2, #1
     SUBS R4, R4, #1
     EOR R5, R5, R5
     BNE L1
     MOV R3, #3
     MOV R4, #4
```

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1B) Discuss how the ARM assembler manages constants using literal pools. Evaluate why the literal pool (and associated instructions like LDR Rd, =const) is necessary for loading large or complex constants and discuss the role of the LTOrg directive in memory management. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 1C) What is the role of the DCB, DCW, and DCD directives in ARM7 assembly? Explain how they are used to declare and initialize data in memory. (3)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 2A) Describe the working of a prioritized simple interrupt handling scheme in an ARM7 processor with the help of a neat flow diagram. (4)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 2B) Discuss the sequence of events that occur when an IRQ/FIQ exception is raised when the processor is in user mode, and explain the ARM instructions that can be used to enable or disable the IRQ/FIQ exceptions. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 2C) With a neat block diagram, explain the datapath activity for the branch instruction in an ARM7 processor. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3A) Draw the 32-bit format of ARM data transfer instructions and discuss in detail the significance of each bit in it. (4)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 3B) Write an assembly language program for ARM to calculate the sum of five words: 1000000000, 2000000000, 3000000000, 4000000000, and 4100000000. (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 3C) Pre-conditions: (3)
R0=0x40000000, R1=0x12345678, R2=0xAABBCCDD, R3=0x1A2A3B4C, R4=0x12345678, R5=0x0000F00, R6=0x45129860, R7=0x88772344, and cpsr=nzCvquFt_USER;
Describe the function of the following ARM7 instructions in detail. Assuming the same given pre-conditions for all instructions, write the post-conditions of the affected registers and flags for each instruction.
i) SMLALNE R0, R1, R2, R3
ii) TSTMI R4, R5
iii) ASR R6, R7, #6

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 4A) Write the shortest possible ARM assembly program to convert 0x76 from the packed BCD number to ASCII, and place the ASCII codes into the R1 and R2 registers. (4)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 4B) Pre-conditions: R14=0x00000000, R13=0x80000000 (3)

Line	Address	Mnemonic	Operands
1	0x00009000	LDR	R1, =0x0000F001
2	0x00009004	BLX	R1
3	0x00009008	MOV	R1, #0x25
4	0x0000900C	MOV	R2, #0x35
5	0x00009010	MOV	R3, #0x45
6	0x00009014	HERE: B	HERE
7	0x0000F000	L1: SUB	R13, R13, #4
8	??	STR	R1, [R13]
9	??	BX	R14

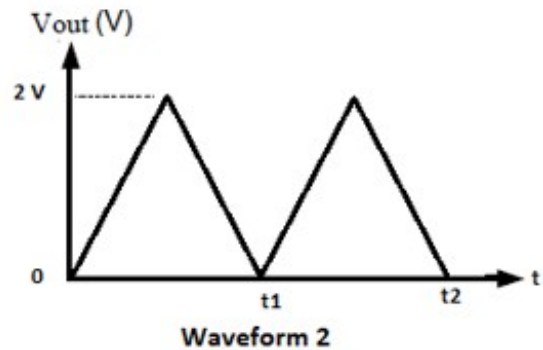
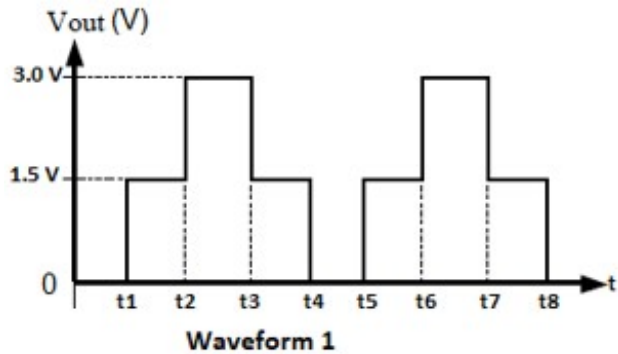
With the given pre-conditions for ARM7 TDMI, compute the following and explain the reasons:

- The value of PC, LR just after completing the execution of "BLX R1" instruction at line 2.
- The memory location addresses of "STR R1, [R13]" and "BX R14" instructions saved at line 8 and 9.
- The value of SP, LR just after completing the execution of "BX R14" instruction at line 9.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 4C) What is a stack in ARM? Explain the four stack structures and the associated STM and LDM instructions available to serve the PUSH and POP operations of the stack in each structure. (3)

- 5A) Interface two switches (SW0 and SW1) to the LPC2148 microcontroller using port lines P0.6 and P0.7 of PORT0, (4)
 respectively. Write an embedded C program to generate the following waveforms using the internal DAC (assume $V_{ref} = 3V$) under the given conditions:
 i) If SW0 is closed, generate waveform 1.
 ii) If SW1 is closed, generate waveform 2.



Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 5B) What is the role of IOPIN, IODIR, IOSET, and IOCLR registers in LPC2148, and explain how to (3)
 configure them to pass logic '1' followed by logic '0' through P0.15 pin.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 5C) For the generic ARM chip, compute the following: (3)
 i) The amount of memory (in bytes) given to the address range of 0x20000000 - 0x200FFFFF for the SRAM bit-addressable region.
 ii) The amount of memory (in bytes) given to the address range of 0x22000000 - 0x23FFFFFF for alias addresses of bit-addressable SRAM
 iii) The bit-alias address of bit 3 of 0x20000004

Course Outcome: "CO3" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_3122:MICROPROCESSORS

#	Course Outcome	Description	Marks
1	1	CO 1	20
2	2	CO 2	7
3	3	CO 3	9
4	4	CO 4	10
5	5	CO 5	4
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_3122:MICROPROCESSORS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	10(20%)
3	Application	5	15(30%)
4	Analysis	7	25(50%)

5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)